DataBase Lab 05 (In-Lab Task)

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Task 1

Welcome Page | hr

Worksheet | Query Builder

```
SELECT SUM(salary) AS total_salary
FROM employees;
```

Query Result

SQL | All Rows Fetched: 1 in 0.001 seconds

	TOTAL_SALARY
1	691716

Task 2

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Worksheet | Query Builder

```
SELECT department_id, AVG(salary) AS avg_salary
FROM employees
GROUP BY department_id;
```

Query Result

SQL | All Rows Fetched: 12 in 0.003 seconds

	DEPARTMENT_ID	AVG_SALARY
1	100	8601.3333333333333333333333333333333333
2	30	4150
3	(null)	7000
4	90	19433.3333333333333333333333333333333333
5	20	9500
6	70	10000
7	110	10154
8	50	3475.5555555555555555555555555555555556
9	80	8955.8823529411764705882352941176470588 24
10	40	6500
11	60	5760
12	10	4400

Task 3

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Worksheet | Query Builder

```
SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id;
```

Query Result

SQL | All Rows Fetched: 12 in 0.002 seconds

	DEPARTMENT_ID	TOTAL_EMPLOYEES
1	100	6
2	30	6
3	(null)	2
4	90	3
5	20	2
6	70	1
7	110	2
8	50	45
9	80	34
10	40	1
11	60	5
12	10	1

Task 4

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Worksheet | Query Builder

```
SELECT job_id, MIN(salary) AS min_salary,
       MAX(salary) AS max_salary
FROM employees
GROUP BY job_id;
```

Query Result

SQL | All Rows Fetched: 19 in 0.002 seconds

	JOB_ID	MIN_SALARY	MAX_SALARY
1	IT_PROG	4200	9000
2	AC_MGR	12008	12008
3	AC_ACCOUNT	8300	8300
4	ST_MAN	5800	8200
5	PU_MAN	11000	11000
6	AD_ASST	4400	4400
7	AD_VP	17000	17000
8	SH_CLERK	2500	4200
9	FI_ACCOUNT	6900	9000
10	FI_MGR	12008	12008
11	PU_CLERK	2500	3100
12	SA_MAN	10500	14000
13	MK_MAN	13000	13000
14	PR_REP	10000	10000
15	AD_PRES	24300	24300
16	SA_REP	6100	11500
17	MK_REP	6000	6000
18	ST_CLERK	2100	3600
19	HR_REP	6500	6500

Task 5

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Worksheet | Query Builder

```
SELECT COUNT(*) AS total_employees
FROM employees
WHERE salary > 50000;
```

Query Result

SQL | All Rows Fetched: 1 in 0.002 seconds

	TOTAL_EMPLOYEES
1	0

Task 6

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Worksheet | Query Builder

```
SELECT department_id, SUM(salary) AS total_salary
FROM employees
GROUP BY department_id
HAVING SUM(salary) > 100000;
```

Query Result

SQL | All Rows Fetched: 2 in 0.002 seconds

	DEPARTMENT_ID	TOTAL_SALARY
1	50	156400
2	80	304500

Task 7

Welcome Page | hr

Worksheet | Query Builder

```
SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id;
```

Query Result

SQL | All Rows Fetched: 12 in 0.002 seconds

	DEPARTMENT_ID	TOTAL_EMPLOYEES
1	100	6
2	30	6
3	(null)	2
4	90	3
5	20	2
6	70	1
7	110	2
8	50	45
9	80	34
10	40	1
11	60	5
12	10	1

Task 8

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Worksheet | Query Builder

```
SELECT department_id, SUM(salary) AS total_salary
FROM employees
GROUP BY department_id
HAVING SUM(salary) > 150000;
```

Query Result

SQL | All Rows Fetched: 2 in 0.002 seconds

	DEPARTMENT_ID	TOTAL_SALARY
1	50	156400
2	80	304500

Task 9

Welcome Page | hr

Worksheet | Query Builder

```
SELECT d.department_name, AVG(e.salary) AS avg_salary
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
WHERE d.department_name LIKE 'S%'
GROUP BY d.department_name;
```

Script Output | Query Result

SQL | All Rows Fetched: 2 in 0.005 seconds

	DEPARTMENT_NAME	AVG_SALARY
1	Shipping	3475.5555555555555555555555555555555555556
2	Sales	8955.8823529411764705882352941176470588 24

Task 10

Worksheet | Query Builder

```
INSERT INTO employees
(employee_id, first_name, last_name, email, phone_number, hire_date, job_id, salary, department_id)
VALUES
(209, 'Mujtaba', 'Qureshi', 'MUJ207', '03001234567', SYSDATE, 'IT_PROG', 60000, 60);
```

Query Result | Script Output

Task completed in 0.026 seconds

```
1 row inserted.
```